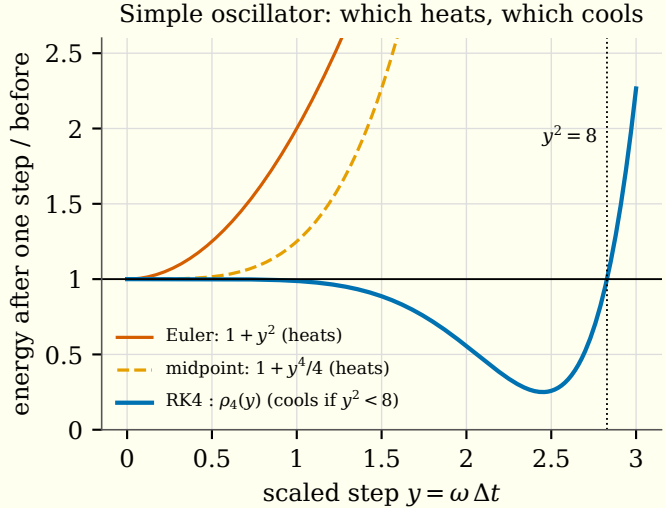
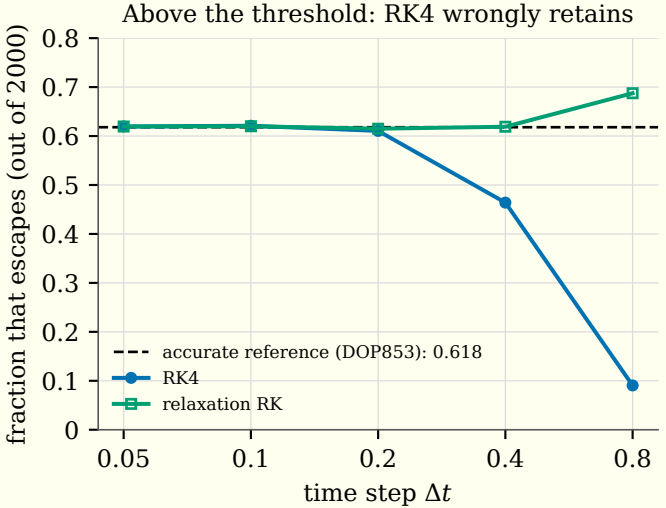

Simple oscillator: which heats, which cools
energy after one step / before
scaled step $y = \omega\,\Delta t$
$y^2 = 8$
Euler: $1 + y^2$ (heats)
midpoint: $1 + y^4/4$ (heats)
RK4 : $\rho_4(y)$ (cools if $y^2 < 8$)
0
0.5
1
1.5
2
2.5
3


Above the threshold: RK4 wrongly retains
fraction that escapes (out of 2000)
time step Δt
accurate reference (DOP853): 0.618
RK4
relaxation RK
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.05
0.1
0.2
0.4
0.8